



INSTITUTIONAL SYSTEMS & HEALTHCARE INFRASTRUCTURE

LLOYDS MEDICAL OS
(LMOS-PNG)

A 100% offline-first, portable clinical operating system designed for remote mining concessions, community health posts, and rural emergency stations. Includes live telemetry vitals, Manchester triage, POS pharmacy inventory, financial shift reconciliation, and tamper-proof Excel archives.

PREPARED FOR	SYSTEM ARCHITECTURE	RELEASE VERSION
Executive Leadership & Mine Medical Board	100% Offline SQLite + Multi-Device LAN	v2.6-PNG Institutional Edition

01 EXECUTIVE OVERVIEW & OPERATING MISSION

STRATEGIC CONTEXT

Remote mining concessions and extraction sites in Papua New Guinea operate under unique environmental and logistical constraints: erratic or non-existent internet connectivity, rugged field conditions, high turnover of community health workers with limited IT training, and significant operational risks regarding pharmaceutical inventory leakage and unrecorded cash transactions.

THE OPERATIONAL CHALLENGE

Standard hospital management platforms (SaaS/Cloud systems) fail in rural mining concessions because they require continuous broadband, complex server installations, and sophisticated operator training. When internet fails, clinics resort to paper ledgers, causing loss of patient clinical histories, stock pilferage, and zero accountability for shift revenues.

The **Lloyds Medical Operating System (LMOS-PNG)** addresses this challenge with a revolutionary **"Zero-Cloud, 1-Click Plug-and-Play"** architecture:

100%
OFFLINE

Self-Contained
SQLite

Runs directly off a USB pen drive or local clinic PC with zero external database server or internet required. Fast, durable, and resilient to sudden power outages.

PSYCHOLOGICAL
UI

Zero-Training
Adoption

Engineered with high-contrast HUD telemetry, CRT-style ECG waveforms, and large, color-coded visual buttons so local community health workers master the workflow in under 15 minutes.

FINANCIAL
AUDIT

Tamper-Proof
Accounting

Daily cash drawer balancing with variance calculation, triple sign-off authorization sheets, and AES-encrypted Excel exports with hidden Admin PIN protection.

02 CORE SYSTEM ARCHITECTURE & SYNC PROTOCOL

INFRASTRUCTURE

The system is designed with a **three-tier deployment model**, allowing flexibility across single-laptop bush stations up to full mining base hospitals:

DEPLOYMENT MODEL	HARDWARE REQUIREMENT	NETWORK REQUIREMENT	BEST SUITED FOR
1. Portable Pen Drive (USB)	Any standard Windows or Mac PC	Zero (100% Offline)	Remote exploration camps, mobile drill-rig triage posts, and bush patrol clinics.
2. Clinic Local Wi-Fi (LAN)	1 host laptop + local Wi-Fi router	Local clinic Wi-Fi (No Internet needed)	Multi-room base hospitals where Triage Nurse, Doctor, and Pharmacist use separate tablets/PCs.
3. Cloud VPS (Optional)	Dedicated Linux VM (AWS / Docker)	Broadband	Centralized head-office oversight across multiple concessions.

HYBRID CLOUD & PEN DRIVE SYNC LOGIC

How Data Syncs When Connecting:

1. **On the Pen Drive:** Every visit, prescription, and fee writes in real time to `server/data/hospital.db`. When removing the drive, clicking "Safe Pen Drive Eject" automatically flushes SQLite WAL buffers and downloads an encrypted Excel snapshot.
2. **Google Sheets & Cloud Sync:** The system maintains an offline buffer. When the clinic laptop detects camp Wi-Fi or a smartphone 4G hotspot, it automatically syncs patient footfall and financial totals to the corporate Google Sheet via background webhook.

03 MODULE A: COMMAND CENTER HUD & LIVE VITALS

REAL-TIME MONITORING

The primary landing screen functions as an institutional Medical Command Center. Designed with an obsidian dark theme, subtle 32px telemetry grid, and high-visibility digital readouts, it presents clinical status at a single glance:

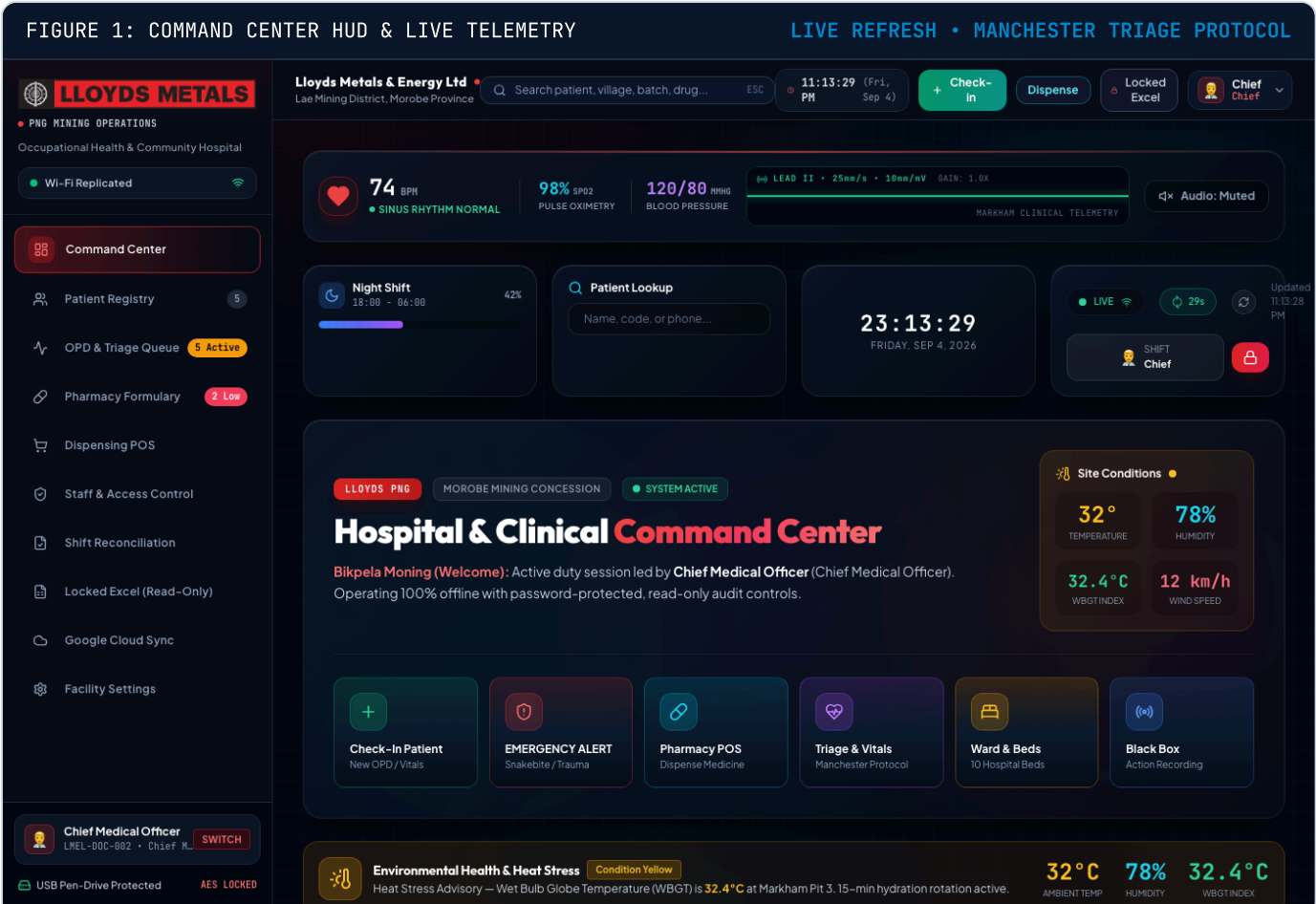


Figure 1: High-contrast Command Center dashboard featuring live ECG waveform monitor (Lead II, 25mm/s), Manchester triage vital signs grid, Quick Check-in launcher, and Emergency Response Trauma Protocols.

1. Real-Time ECG Heart Monitor

Features CRT scanline overlay, phosphor persistence trails, 60–100 BPM heart rate calculation, and an optional audible systolic beep toggle for trauma resuscitation bays.

2. Manchester Triage Vitals Tile

Displays Heart Rate, Blood Pressure (120/80 mmHg), SpO2 Oxygen Saturation, Body Temperature, and Respiration Rate with immediate color-coded alert badges (Normal / Warning / Critical).

3. Emergency Response Timer

Built-in 1-click clinical timers with pre-loaded treatment protocols for: *Cerebral Malaria Coma*, *Death Adder Snakebite*, *Cardiac Arrest*, *Blast Trauma*, and *Severe Heat Dehydration*.

4. Black Box Action Recorder

Under-the-hood immutable telemetry logger that timestamps every triage admission, dispensation, dosage change, and cash entry for medico-legal protection.

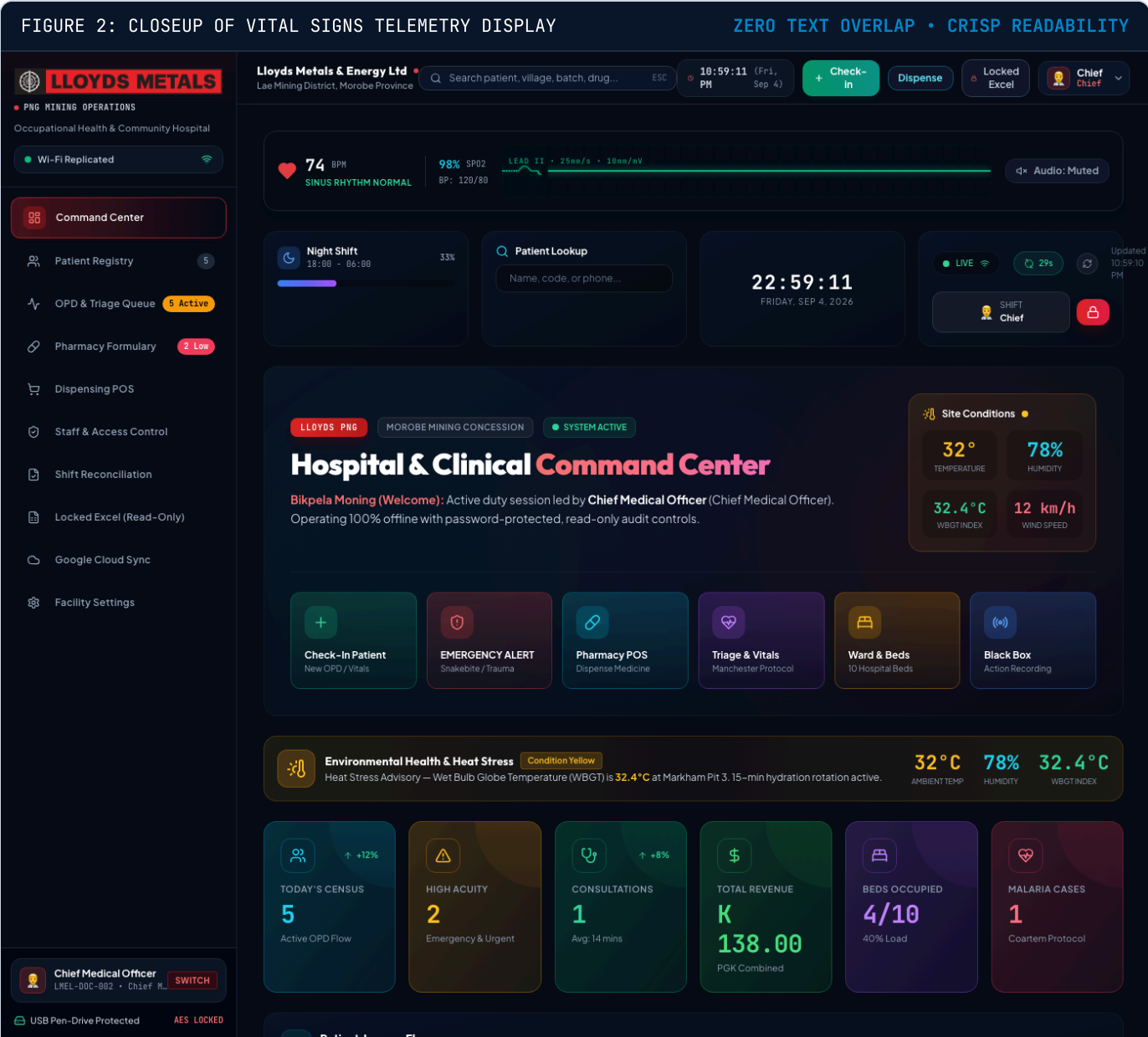


Figure 2: Verified vital signs telemetry tile showing crisp typography for Blood Pressure (120/80 mmHg), SpO2 (98%), Heart Rate (72 bpm), and Respiratory Rate (16/min).

04 MODULE B: OUTPATIENT DEPARTMENT (OPD) & TRIAGE

PATIENT FLOW

The Outpatient Department (OPD) queue coordinates the entire patient journey from triage arrival through doctor consultation and pharmacy dispatch:

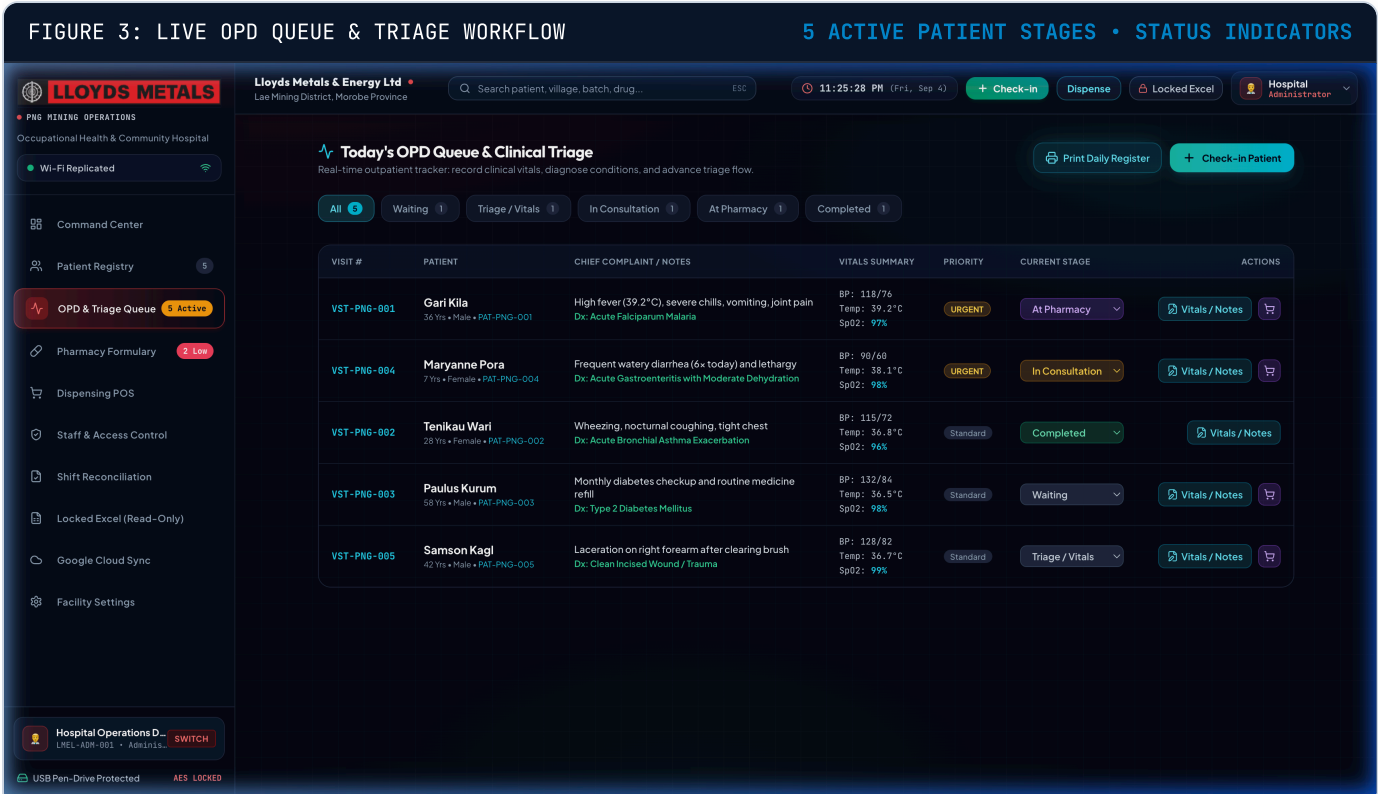


Figure 3: OPD clinical queue showing live status tabs (Waiting, Triage, In Consultation, At Pharmacy, Completed), patient triage severity, vital signs, chief complaints, and action buttons.

Token Management & Patient Intake

Instant registration assigns unique institutional codes (e.g. PAT-PNG-005) with blood group, known drug allergies (Penicillin, Sulphonamides), and emergency family contacts.

Digital Health Pass & Card Generation

1-click creation of Patient ID Cards with scannable QR codes for repeat visits, eliminating paper file misplacement and speeding up intake in rural stations.

05 MODULE C: PHARMACY FORMULARY & AUTOMATED DISPENSING POS

INVENTORY CONTROL

The Pharmacy module enforces strict inventory control calibrated to Papua New Guinea Essential Medicines:

- **Formulary Pre-loads:** Antimalarials (Artemether-Lumefantrine / Coartem), Antibiotics (Amoxicillin, Ciprofloxacin), Analgesics (Paracetamol, Ibuprofen), Rehydration (WHO ORS), and Respiratory Inhalers (Salbutamol).
- **Real-Time Stock Depletion:** Dispensing directly decrements physical inventory and alerts staff when stock falls below reorder thresholds (e.g. Zinc Sulfate at 8 units remaining).
- **Automated Official Tax Receipts:** Instant generation of bilingual receipts with unit prices, concession subsidies, and dispensing officer signatures.

ANTI-THEFT MEDICATION TRACKING

Every unit of medication dispensed requires a linked patient ID and visit record. Unlinked dispensations are impossible, completely preventing off-the-record medicine sales and loss of vital supplies in remote camps.

06 MODULE D: END-OF-DAY SHIFT CLOSEOUT & FINANCIAL AUDIT

FINANCIAL
RECONCILIATION

At the conclusion of each 24-hour cycle or duty shift, staff execute the End-of-Day Balancing procedure. The system tallies total consultation fees, prescription sales, and compares system-expected revenue against the physically counted cash in the clinic safe:

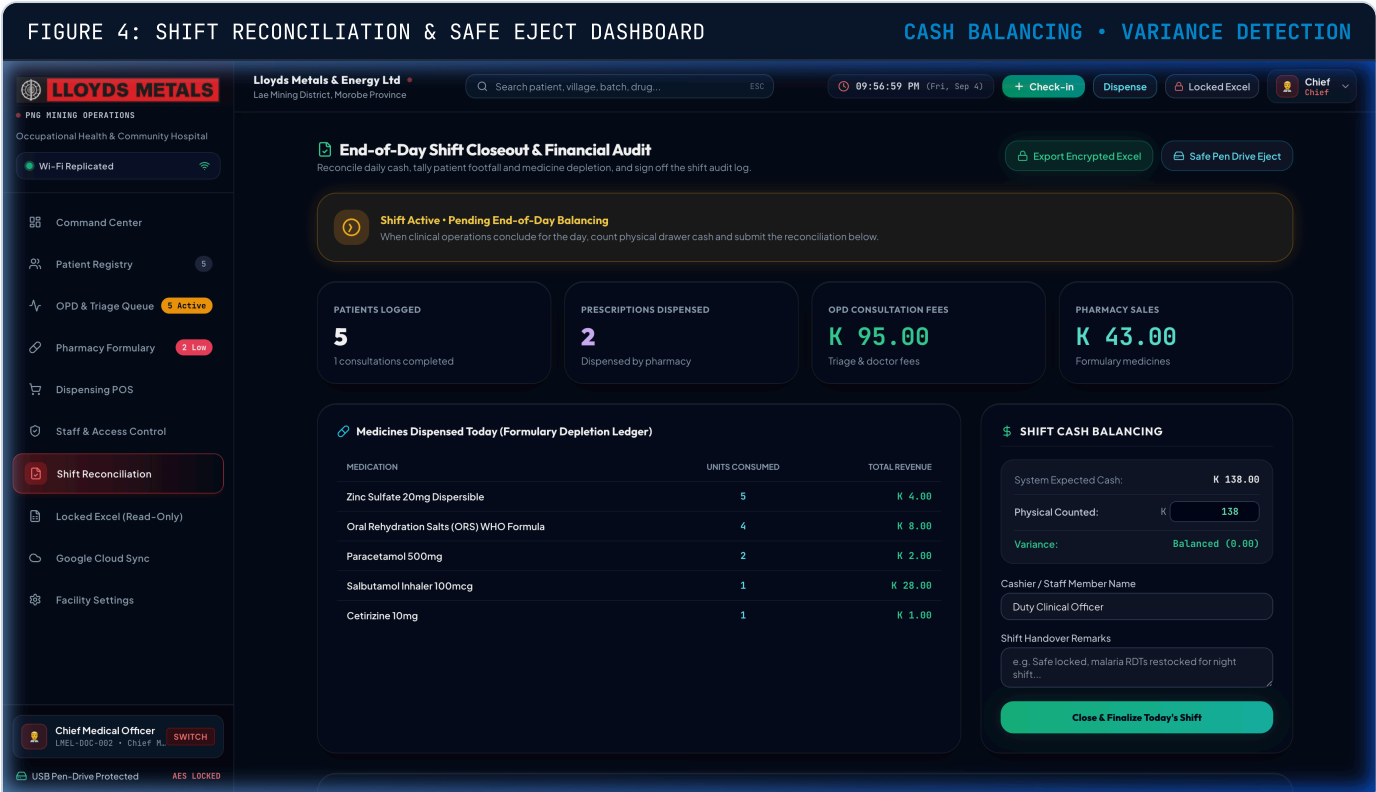


Figure 4: End-of-Day reconciliation view showing system cash tally (K 138.00), drawer count, variance calculator, formulary consumption tally, and Safe Pen Drive Eject.

07 MODULE E: INSTITUTIONAL PRINTABLE REPORTS & PDF ENGINE

AUDIT
DOCUMENTATION

All reports print with crisp institutional formatting designed to satisfy internal mine auditors and Papua New Guinea health authority inspectors:

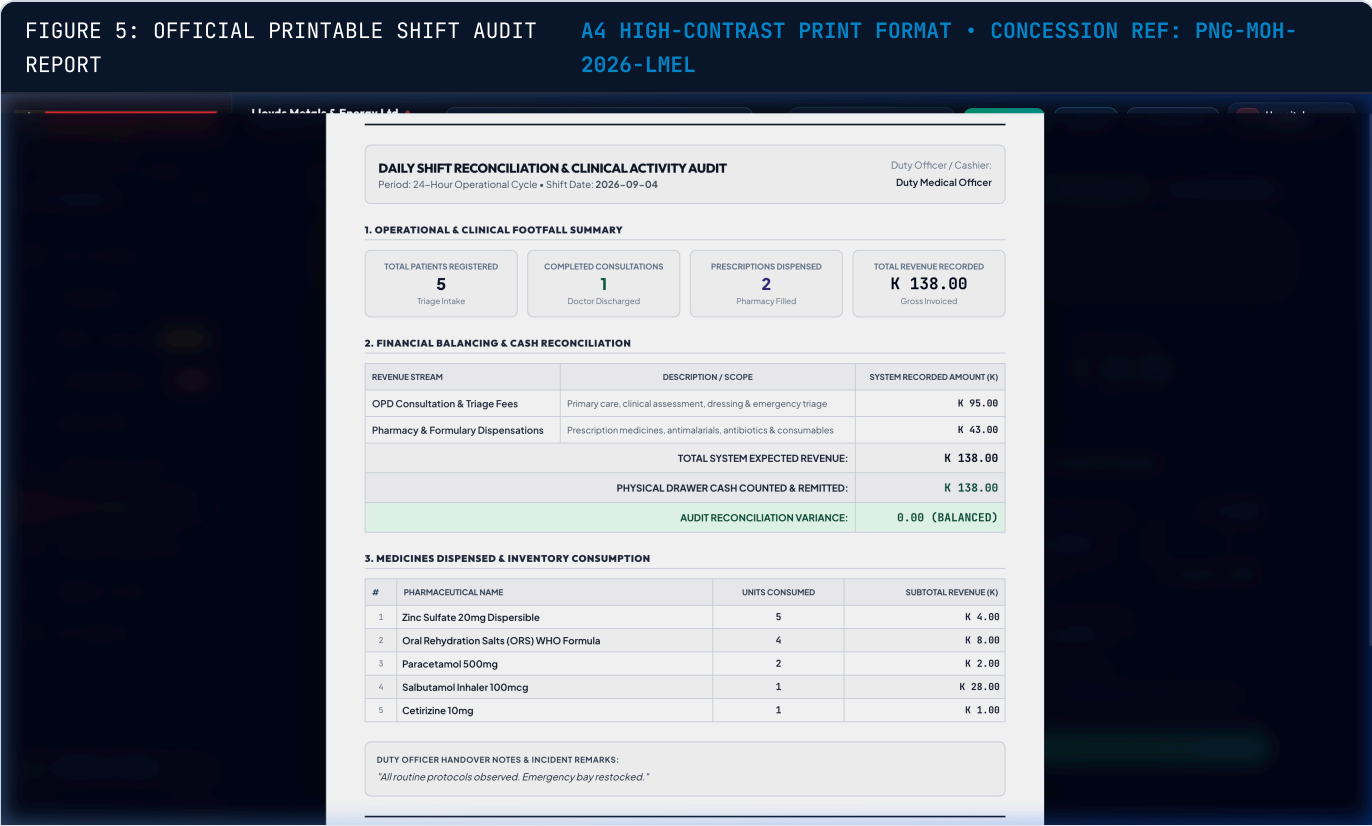


Figure 5: Official Daily Shift Reconciliation & Clinical Activity Audit Sheet showing executive footfall summary, financial breakdown (OPD fees + pharmacy sales), and inventory depletion table.



08 STEP-BY-STEP SETUP, DEPLOYMENT & OPERATING GUIDE OPERATIONAL RUNBOOK

This section details exact operating instructions so that any clinic officer or technician can launch, operate, and maintain the system with zero ambiguity.

A. Quick Launch from USB Pen Drive (Zero Installation)

1. Insert the Lloyds Medical OS USB Pen Drive into any laptop or desktop computer at the clinic.
2. Open the drive folder:
 - **On Windows:** Double-click `Start_Hospital_Windows.bat`
 - **On macOS:** Double-click `Start_Hospital_Mac.command`
3. A terminal window opens automatically and launches the local offline engine.
4. Open Google Chrome or any web browser and navigate to: <http://localhost:4000>
5. The complete Lloyds Medical Command Center loads instantly. Zero internet is needed!

B. Clinic Multi-Device Wi-Fi Network (LAN Mode)

1. Connect the primary clinic laptop to the clinic Wi-Fi router (internet is NOT required; the local Wi-Fi acts as a local bridge).
2. Launch the server on the primary laptop. Note the LAN address displayed in the startup screen (e.g. `http://10.130.98.192:4000`).
3. Any tablet, doctor laptop, or pharmacy computer connected to that same Wi-Fi can now open that address in their browser to access the live clinic database simultaneously.

C. Master Access Credentials & PINs

ACCOUNT / ROLE	USERNAME	DEFAULT PASSWORD	ACCESS PRIVILEGES
Master Admin Console	admin	lloyds2026	Full system control, staff management, password changes, and hospital settings.
Hidden Excel Admin PIN	(Unlock PIN)	lloyds2026	Reveals export password and allows generating encrypted Excel archives.
Supervising Medical Officer	doctor	lloyds2026	Triage diagnosis, clinical prescribing, doctor notes, and shift sign-off.
Senior Triage Nurse	triage_officer	lloyds2026	Patient check-in, vitals logging, emergency timers, and queue updates.

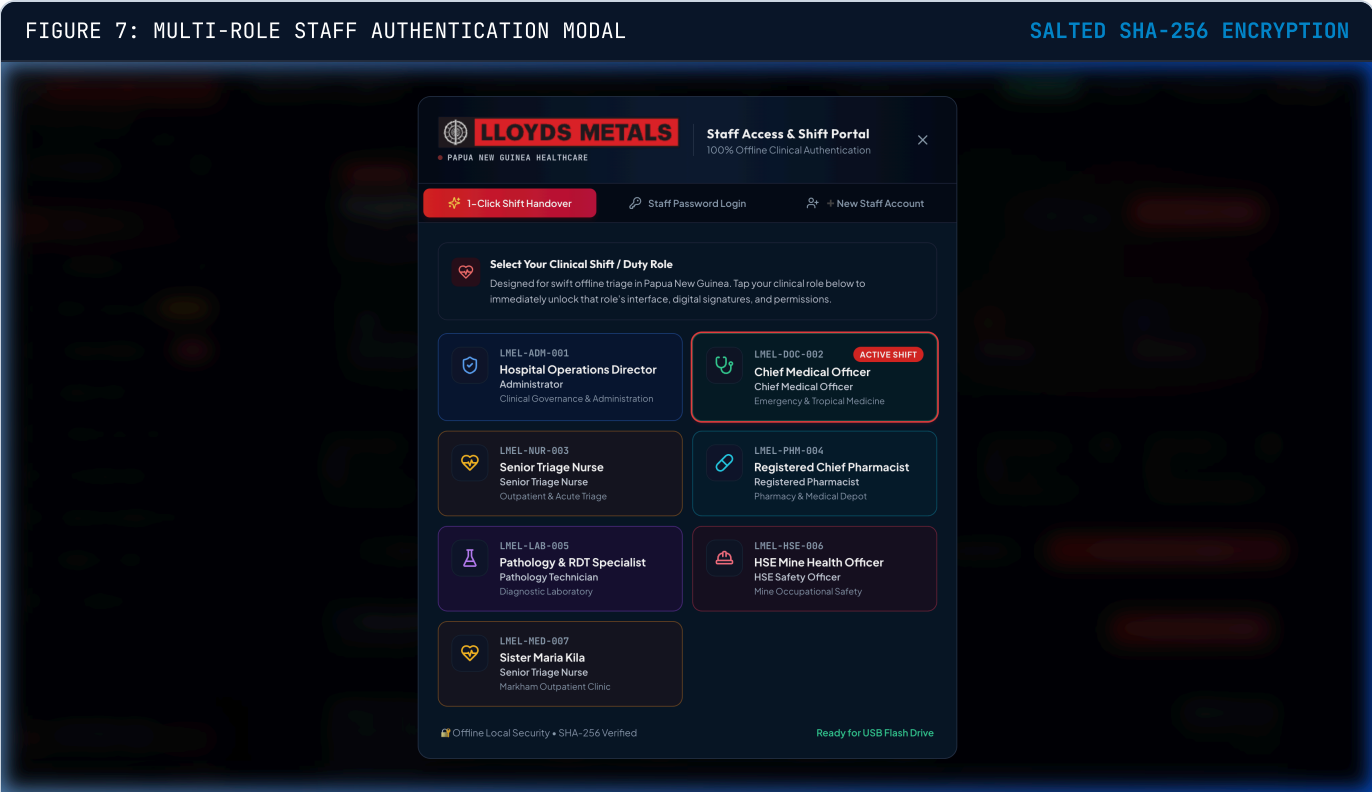


Figure 7: Secure offline staff login modal supporting instant account creation, multi-role assignment (Doctor, Nurse, Pharmacist), and salted SHA-256 password protection.

09

STANDARD DAILY CLINIC OPERATING PROCEDURE (SOP)

STANDARD OPERATING PROCEDURE

STEP 1: MORNING SHIFT OPENING

- 1. Power on the primary clinic PC.
- 2. Launch `Start_Hospital_Windows.bat`.
- 3. Log into Command Center.
- 4. Verify Pharmacy low-stock alerts and temperature monitors.

STEP 2: PATIENT CHECK-IN & TRIAGE

- 1. Click "Quick Check-in" for new or returning worker.
- 2. Enter Vital Signs (BP, HR, SpO2, Temp).
- 3. Assign Manchester Priority (Immediate / Urgent / Standard).
- 4. Patient automatically advances to Doctor Queue.

STEP 3: DOCTOR CONSULTATION

- 1. Open OPD Queue and select patient.
- 2. Record clinical diagnosis and physician notes.
- 3. Click "Save & Send to Pharmacy".
- 4. Prescriptions flow seamlessly to the dispensary.

STEP 4: PHARMACY DISPENSATION

- 1. Select patient in Dispense POS.
- 2. Verify dosage and stock availability.
- 3. Collect consultation and medicine fee.
- 4. Print official medical invoice receipt (A4 or thermal).

STEP 5: EVENING SHIFT CLOSEOUT & CASH BALANCING

- 1. Open "End-of-Day" tab.
- 2. Count physical cash in drawer and enter total.
- 3. Verify variance displays **0.00 BALANCED**.
- 4. Enter handover notes and click "Close & Finalize Today's Shift".
- 5. Click "**Print Shift Audit Report**", sign and affix official concession stamp.
- 6. Click "**Safe Pen Drive Eject**" to safely sync and remove the drive.

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TECHNICAL SPECIFICATIONS & DATA GOVERNANCE

SYSTEM SPECS

SPECIFICATION COMPONENT	IMPLEMENTATION DETAIL
Primary Database	Embedded SQLite 3.x (ACID Compliant, WAL Journaling, zero background daemon).
Database File Location	<code>server/data/hospital.db</code> (Self-contained, portable, single-file archive).
Frontend Architecture	React 18 + Vite + Tailwind CSS + Canvas telemetry HUD monitors.
Backend Engine	Node.js Express microservice with local RESTful endpoints.
Cryptographic Security	Salted SHA-256 staff password hashing; AES-128/256 Excel spreadsheet encryption.
Print Standardization	High-fidelity CSS print engine conforming to ISO A4 (210 x 297 mm) and 80mm POS thermal roll.
Regulatory Registration	Registered Concession Clinic ID: PNG-M0H-LMEL-2026 (Papua New Guinea).